Gawtam Chithra Ramesh

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Summary

Robotics and machine-learning researcher (MSc Systems, Control & Robotics, KTH) working at the intersection of generative models, reinforcement learning, and end-to-end perception-to-planning for embodied agents. My Master's thesis develops inference-time guidance for a learned flow-matching motion prior; prior work spans deep and multi-agent reinforcement learning (PPO), self-supervised and generative deep learning, and autonomous-vehicle motion planning and control. Three lead-author publications. I aim to advance closed-loop, world-model-based end-to-end learning for automated driving.

Education

KTH Royal Institute of Technology — Aug. 2024 – Jul. 2026
Master of Science in Systems, Controls, and Robotics - 4.62 / 5.0 — *Stockholm, Sweden*

Indian Institute of Technology Madras — Aug. 2019 – Jul. 2024
Dual Degree (B.Tech in Engineering Design and M.Tech in Robotics) - 7.9 / 10 — *Chennai, India*

Publications

- **G. Chithra Ramesh**, et al. "Robust Object-layer Construction of 3D Scene Graphs Using Instance Segmentation." *International Conference on Image Processing and Vision Engineering (IMPROVE 2026).* (Accepted, Long Oral, Nominated for Best Paper Award)
- **G. Chithra Ramesh**, et al. "Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction." *International Conference on Pattern Recognition (ICPR 2026).* (Under Review)
- **G. Chithra Ramesh**, D. Blanco-Mulero, Y. Dong, J. Borràs Sol, C. Torras, F. T. Pokorny. "Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided Vision-Language Models." *IROS 2025 Workshop on Robotic Manipulation of Deformable Objects (ROMADO).* (Accepted)

Scholastic Achievements

- Awarded (among 30 of 800+) the IIT Madras Young Research Fellowship based on remarkable research potential.
- Secured 97.64 Percentile in JEE Advanced & 99.10 Percentile in JEE Mains out of 1.5 Million across India.
- Secured an All India Rank of 52 out of 50k in Undergraduate Common Entrance Examination for Design.

Professional Experience

Master's Thesis on Generative Planning for Long-Horizon Manipulation — (Ongoing) Jan 2026 - Jun 2026
ABB Robotics, Mentors - Matthew Lock and Jonathan Styrud — *Västerås, Sweden*

- Designing an inference-time guidance method for a pretrained flow-matching trajectory planner, allowing a single learned motion prior to satisfy arbitrary task specifications without retraining or per-task expert demonstrations.
- Using Signal Temporal Logic with STLCG++ as the specification language, enabling differentiable robustness gradients to steer generation toward compositional, temporally-extended objectives.
- Validating on Maze2D and OGBench long-horizon benchmarks; benchmarking head-to-head against state-of-the-art STL planners (DAG-STL, TeLoGraF).
- Collaborating with ABB Robotics R&D on transfer of the approach to long-horizon 6-DoF manipulation use cases.

Device Research Intern on 3D Scene Graph — Jul 2025 - Dec 2025
Ericsson Research, Mentor - Puren Guler and Hector Caltenco — *Lund, Sweden*

- First-author paper accepted as Long Oral and nominated for Best Paper Award at **IMPROVE 2026**; second first-author paper under review at **ICPR 2026**.
- Developed a robust 3DSG pipeline by integrating instance segmentation and tracking into a SOTA framework (Hydra), reducing sensitivity to parameter tuning across diverse environments.
- Designed a synchronized RGB-D inpainting module for privacy-aware 3DSG construction, maintaining consistency across RGB and depth channels.
- Analyzed 3DSG frameworks to identify research gaps in object merging, parameter sensitivity, and privacy protection.
- Benchmarked SOTA inpainting algorithms to balance reconstruction fidelity with real-time performance.

Graduate Robotics Intern on Modular Collaborative Robots — Dec 2022 - Jul 2023
Systemantics India Pvt. Ltd., Mentor - Jagannath Raju — *Bangalore, India*

- Developed backward-chained Behavior Trees for 6DOF manipulator to enable robust, long-horizon trajectory planning
- Designed and implemented a 6-DOF trajectory generation and control pipeline in C++, achieving real-time joint actuation via low-latency communication with motor controllers using D-Bus and SocketCAN.

Tested Multi Object Detection and Tracking Models June 2021 – July 2021
Blurgs Research Labs Pvt. Ltd. *Remote, India*

- Analyzed 5 State of the art Multi object tracking networks for startup's first product for The Indian Navy
- Trained and tested various MOT networks like **Siam-MOT** and **FairMOT** for Surveillance with Drones and CCTV.
- Used **FairMOT** to track and get the trail of multiple moving objects and Re-identify objects in real-time.

Graduate Teaching Assistant Aug 2023 – Nov 2023
Field and Service Robotics (ID4060), Mentor - Bijo Sebastian *IIT Madras, India*

- Involved in design and evaluation of assignments based on ROS and CoppeliaSim for 40 students.
- Held meetings with small groups of students with diverse backgrounds to review their assignments and projects.

Research Experience

Graduate Robotics Research on Deformable Object Manipulation Jan 2025 - Nov 2025
Robotics Perception and Learning, Advisor - Florian T. Pokorny *KTH, Sweden*

- Built prompt–taxonomy framework to generate structured, robot-parsable specifications from high-level prompts, demonstrating a pipeline for translating natural-language instructions into structured outputs for downstream manipulation.
- Performed quantitative analysis to identify failure modes and guide iterative improvements to the taxonomy and prompts
- Authored and submitted the results at ROMADO, an **IROS2025** workshop.

Dual Degree Project on Structure from Motion Jan 2024 - May 2024
INSPIRE Lab, Advisor - Nirav Patel *IIT Madras, India*

- Developed a 3D perception system using COLMAP (Structure from Motion) from monocular RGB images to build cost-efficient human torso reconstruction for more accessible ultrasound.
- Developed an autonomous ultrasound scanning system using ROS, integrating a 5-DoF robotic arm with a single RGB camera to achieve cost-effective 3D perception for medical imaging.
- Implemented motion planning and control using ROS and MoveIt, and simulated the 5DOF manipulation in Gazebo.
- Integrated perception algorithms within ROS, processing 3D point clouds from RGB images to enable accurate trajectory planning for the ultrasound probe.

Offline And Online Motion Planning & Control of Unmanned Aerial Vehicles Sep 2021 - Apr 2022
Young Research Fellowship, Advisor - *Prof. Satadal Ghosh* *IIT Madras, India*

- Developed a Target Driven Motion Planning and Obstacle Avoidance algorithm for UAVs.
- Used Proportional Navigation and Acceleration Velocity obstacles in 2D field to achieve 20% faster and more computer efficient algorithm compared classical aerospace algorithms.

Course Projects

Deep Learning, Advanced: Generative & Self-Supervised Models | DD2610 - KTH Nov 2025 - Jan 2026

- Reimplemented **MeanFlow**, a one-step generative flow-matching model, from scratch in PyTorch; ablated time samplers, classifier-free guidance (ω, κ, p_c), and the $r \neq t$ ratio, reaching best FID ≈160 on CIFAR-10 (epoch 600) and 75% optimal r=t mixing on MNIST – directly informing my thesis on generative trajectory planning.
- Built Masked Autoencoders with a Vision Transformer encoder from scratch (75% patch masking, MSE on masked patches), reducing reconstruction loss from 0.385 to 0.127 over 5 epochs on CIFAR-10.
- Implemented contrastive (**SimCLR**, NT-Xent τ=0.07) and semi-supervised (**FixMatch**, confidence threshold 0.95) learners in JAX with a ResNet-18 backbone, reaching ∼24% linear-probe accuracy from only 1% of labels.

Robot Learning and Embodied AI | DD2600 - KTH Sep 2025 - Oct 2025

- Implemented a 3D semantic object mapping pipeline using RGB-D data from ARKitScenes, integrating OwlV2 for open-vocabulary object detection, SAM for segmentation, and CLIP for semantic grounding.
- Deployed a pre-trained Vision-Language-Action (VLA) model for robot manipulation, executing policy inference from visual inputs and natural-language commands.
- Characterised generalisation and robustness failures of the VLA on unseen tasks, contributing concrete observations on where current VLA architectures break down.

Multi Agent Path Finding in Unity3D | DD2438 - KTH Feb 2025 - May 2025

- Developed Hybrid A* path planner and waypoint tracking system in Unity3D for kinematically constrained single-agent (car/drone) navigation, achieving competitive performance.
- Engineered Conflict-Based Search (CBS) to solve MAPF for over 20 holonomic (drones) and non-holonomic (cars) agents, minimizing makespan in shared environments.
- Implemented a greedy dynamic goal assignment strategy for multi-agent teams ("bakery problem") integrated with CBS, optimizing task allocation and minimizing overall travel time.

Reinforcement Learning for Pacman Capture The Flag | DD2438 - KTH Mar 2025 - May 2025

- Designed a hybrid AI for Pacman CTF (Unity3D), merging a custom Behavior Tree for high level strategy with RL models for specialized actions (attack, defend, escape).
- Trained RL (PPO) policies via Unity ML-Agents with role-specific designs (observations, rewards) and advanced methods (phased training, self-play, curriculum learning) for adaptive agent behavior.
- Evaluated strategic agent decision-making and adaptive behavior switching in a competitive, partially observable, dynamic multi-agent Pacman environment.

Image analysis and Computer vision | DD2423 - KTH Oct 2024 - Nov 2024

- Applied Fourier Transforms for image filtering and frequency analysis; designed Gaussian smoothing filters via FFT to reduce noise and study effects on varying image resolutions.
- Implemented a multi-scale differential geometry edge detector (computing L_{vv}, L_{vvv} for zerocrossings) and used Hough Transform for line detection from edges.
- Implemented image matching (SIFT, RANSAC) to estimate homographies (planar scenes) and fundamental matrices, enabling 3D scene reconstruction via triangulation

Introduction to robotics | DD2410 - KTH Sep 2024 - Oct 2024

- Implemented an iterative inverse kinematics solver from scratch for a 7-DOF KUKA robot arm in ROS
- Applied RRT* to find collision-free paths for a Dubins car, showcasing path planning for non-holonomic vehicles.
- Engineered 2D occupancy grid mapping solutions in ROS, processing laser scans to build maps, identify free space, and generate C-spaces for navigation
- Integrated a Behavior Tree mission planner for a TIAGo mobile manipulator (using ROS and Gazebo), enabling autonomous navigation, vision-based grasping, and robust task execution (e.g., kidnapped robot recovery).

Control of Automotive Systems | ED4060 - IIT Madras Jun 2022 - Nov 2022

- Designed a heading angle controller for autonomous ground vehicle considering its vehicle dynamics & reduced error between the desired heading angle and the actual heading angle while working within given performance specifications
- Improved performance of pneumatic brake system by designing P and PI controllers to meet given specifications.
- Designed a nonlinear controller (Sliding Mode Control) to attenuate the effect of external forces on a tractor's hydraulic hitch, to reduce excessive vibrations and to prevent front-wheel lift-off.

Technical Skills

- **Languages** : Python, C, C++, C#
- **Robotics** : ROS2, Unity3D, MATLAB, MuJoCo
- **DL** : Pytorch, OpenCV, wandb, JAX
- **Tools** : VS Code, Docker, Git, LaTeX, CUDA, Slurm
- **CAD** : Fusion360, Solidworks, Ansys, COMSOL

Relevant Coursework

Robotics - Introduction to Field and Service Robotics, Mechanics of Serial Robots,
Controls - Modelling of Dynamical Systems, Control Theory Advanced & Practice, Controls of Automotive Systems
Programming - Image Analysis and Computer Vision, Fundamentals of Deep Learning, Data Structures and Algorithms